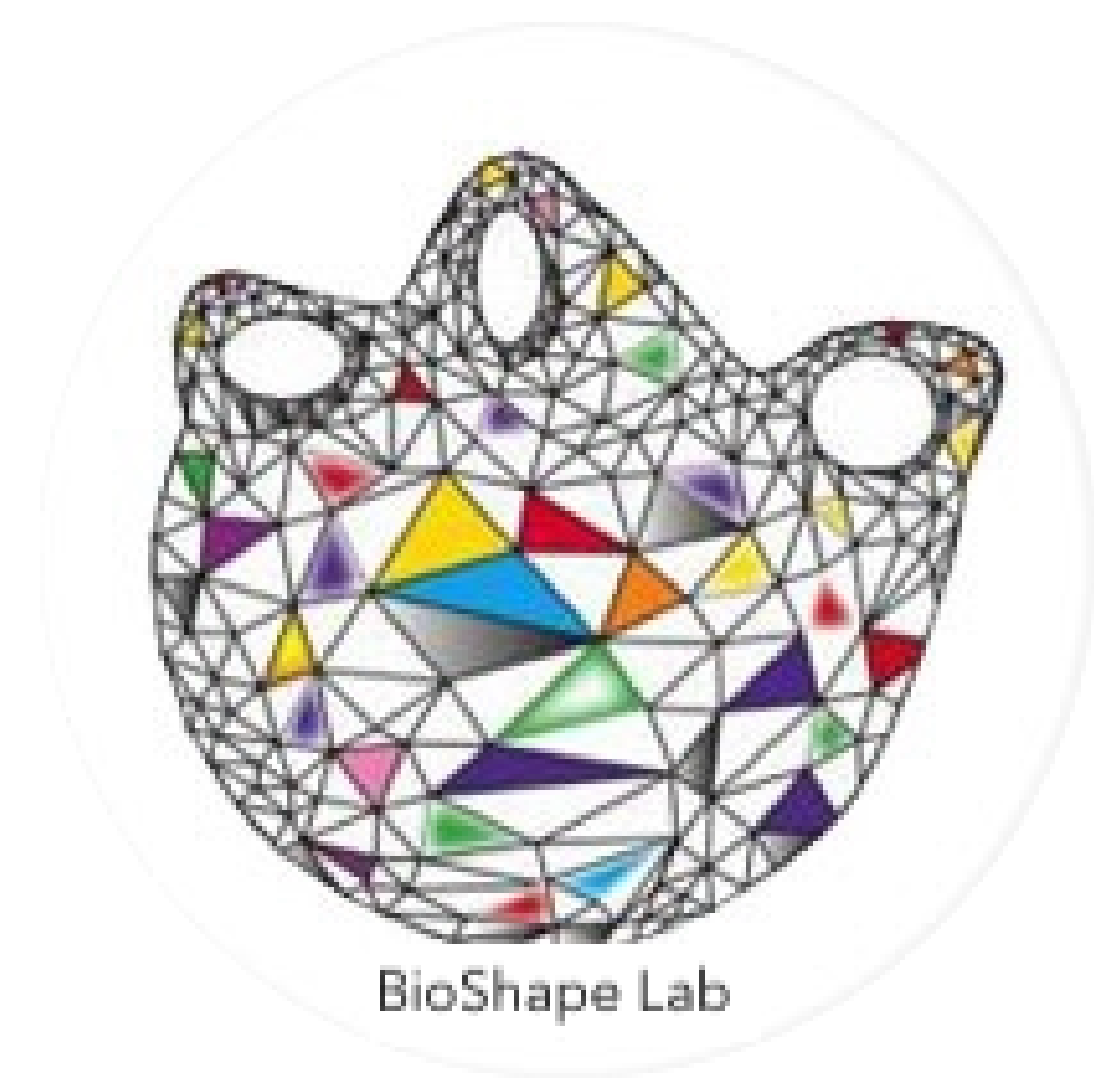


From MultiAgent Simulation to Causal Multilevel Graphs in Parkinson's Disease

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MOTIVATIONS

Although the clinical symptoms of Parkinson's disease are well characterized, its pathological progression involves interacting processes across molecular, intracellular, cellular and extracellular scales. Together, these mechanisms form a complex and partially self-reinforcing cascade.

This complexity motivates an agent-based modelling approach, in which proteins, organelles, neurons and glial cells are represented as heterogeneous autonomous agents with local perception, internal states and rule-based behaviour.

Rather than providing a clinically calibrated predictive model, this project develops an exploratory and explainable framework. Agent-based simulation, causal logging and multilevel graph analysis are combined to connect local events with recurrent higher-level patterns.

CONCLUSIONS

KEY OBSERVATIONS:

Graph contraction reduced the causal representation from **145,818 temporal nodes** to only **19 higher-level patterns**.

The dominant pathological pathway connected **α -synuclein aggregation, neuronal stress, glial activation and persistent inflammation**, suggesting that intracellular and extracellular inflammation amplify one another.

Dopamine output decreased by **33%** over **500 ticks**.

Limitations and future work. The current model remains exploratory and is not clinically calibrated. Future developments should include parameter calibration on biological data, richer stochastic dynamics, spatial validation, systematic comparison of intervention scenarios and a stronger connection between simulated mechanisms and symptom-level outcomes.

REFERENCES

- [1] Caputo M., Russo M. and Merelli E. Space of Data through the Lens of Multilevel Graph
- [2] Wooldridge M. An Introduction to MultiAgent Systems
- [3] Nature Video Understanding Parkinson's disease
- [4] ammermali/Parkinson-Simulation, source code available at:



BIOLOGICAL BACKGROUND: PARKINSON'S DISEASE

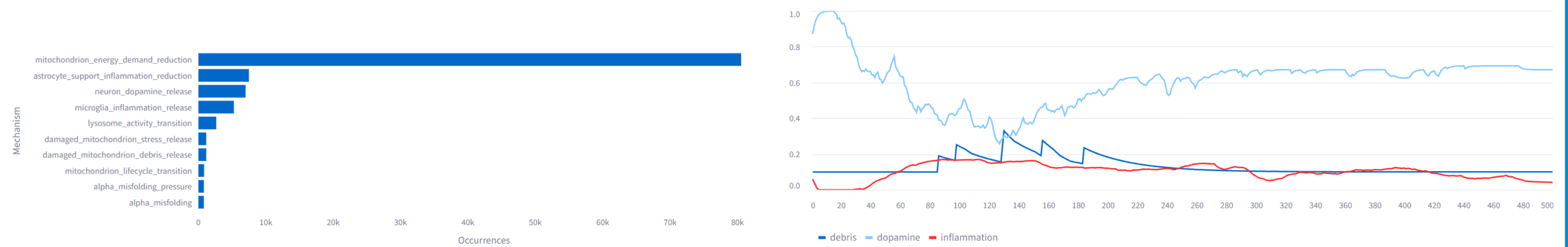
Parkinson's disease is associated with the progressive loss of **dopamine-producing neurons in the Substantia Nigra**. These neurons normally supply dopamine to the striatum, helping regulate movement. As dopamine levels decline, motor control becomes impaired, producing symptoms such as slowed movement, rigidity and tremor.

The mechanisms responsible for neuronal death are not fully understood, but several interacting processes are involved. The protein **α -synuclein can misfold and aggregate**, forming Lewy bodies that could **impair lysosomes and clearance mechanisms**. At the same time, mitochondrial dysfunction reduces cellular energy production and increases oxidative stress. Stressed or dying neurons may release α -synuclein into the extracellular environment. The protein can be taken up by neighbouring cells, where it may promote further aggregation. Cellular damage also **activates microglia and astrocytes**, producing a neuroinflammatory response that may initially be protective but can become harmful when persistent.

OUR CONTRIBUTION

Research question. Can multilevel graph contraction transform detailed agent-based execution traces into compact and interpretable representations of recurrent Parkinsonian mechanisms?

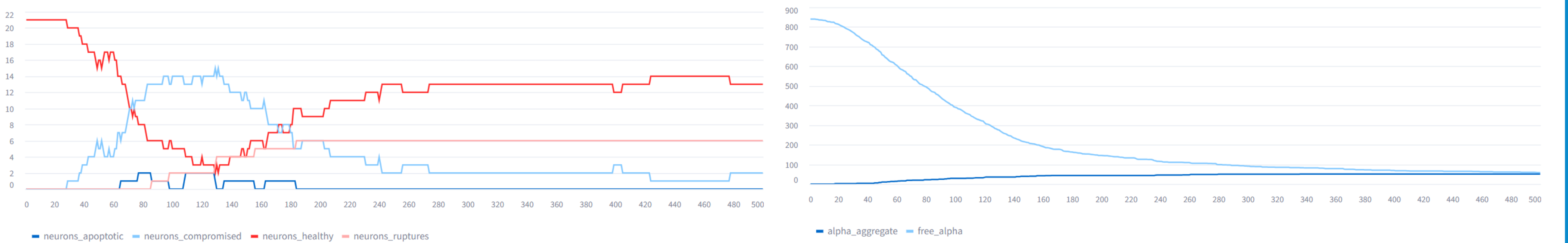
We developed an exploratory multiagent simulation of Parkinsonian neurodegeneration that connects intracellular and extracellular processes within a shared substantia nigra environment.



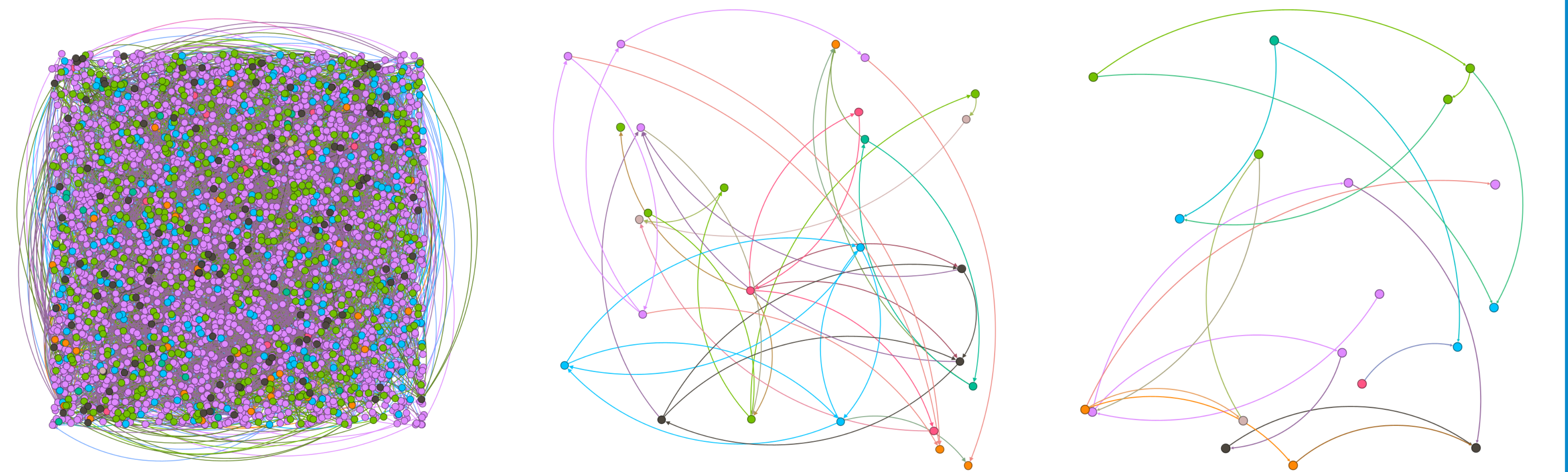
The model represents dopaminergic neurons, α -synuclein, mitochondria, lysosomes, microglia and astrocytes as adaptive agents. Their local interactions generate changes in oxidative stress, protein aggregation, inflammation, extracellular debris, neuronal damage and dopamine output.

Nested biological simulation. Dopaminergic neurons act as macro-agents containing intracellular α -synuclein, mitochondria and lysosomes, while neurons and glial cells interact through the extracellular environment.

Causal event logging. State transitions, perceptions, actions and environmental effects are recorded as a timed causal trace.



Multilevel graph abstraction. The execution trace is progressively transformed from a detailed temporal graph into increasingly compact representations. This pipeline makes it possible to inspect the simulation at multiple scales, from individual biological events to emergent causal mechanisms, while preserving the connection with the lower levels.



G1 - Time-Contraction Graph

G2 - Agent-Clustering Graph

G3 - Topological Motif Graph